

Marlena Bannick

Education

- 2020–2025 **Doctor of Philosophy in Biostatistics**, *University of Washington*.
Thomas R. Fleming Excellence in Biostatistics Award
- 2016–2019 **Master of Science in Biostatistics**, *University of Washington*.
Senior MS in Biostatistics Award
- 2012–2016 **Bachelor of Science in Public Health**, *University of Washington*.
Minor in Mathematics, College Honors, *magna cum laude*, Phi Beta Kappa, Husky 100

Experience

- 09/2025–
present **Biostatistician III, Seattle Children's Therapeutics**, *Seattle Children's*.
 - Biostatistician for an early phase portfolio of CAR T-cell therapy products at Seattle Children's Therapeutics, including for brain tumors, leukemias, various solid tumors, and lupus.
 - Consult on clinical trial design for principal investigators within Seattle Children's Research Institute.
- 05–08/2024 **Biostatistics Graduate Intern, Cell Therapy**, *Bristol Myers Squibb*.
 - Designed and conducted simulations to answer questions related to the design and analysis of clinical trials of CAR-T cell therapy for blood cancers, with a focus on the duration of response to therapy.
 - Summarized and communicated key insights to project stakeholders using engaging presentations.
 - Built R Shiny application to allow others to run simulations for their specific disease area.
- 09/2020–
09/2025 **Graduate Research Assistant**, *Fred Hutchinson Cancer Research Center*.
 - Developed estimators that incorporate real-world data into active-control trials in HIV prevention.
 - Improved the efficiency of existing HIV incidence estimators by augmenting them with information from individual-level data on HIV testing history.
 - Proposing new estimators that correct for selection bias during trial recruitment.
 - Authored a publication on best practices for observational studies of COVID-19 vaccines.
- 09/2019–
09/2020 **Mathematical Sciences Researcher**, *Institute for Health Metrics and Evaluation*.
 - Developed quantitative methods and modeling strategies that incorporated all possible relevant global health data to achieve credible and policy-relevant results.
 - Implemented methods into code that addressed analytical challenges across teams at IHME.
 - Led engineering design and implementation efforts for a disease modeling software program in Python, managing feature requests and expectations from key stakeholders across the institute.
- 09/2016–
09/2019 **Post Bachelor Fellow**, *Institute for Health Metrics and Evaluation*.
 - Designed data processing and statistical modeling software for a large cluster computing platform used by dozens of disease modelers as the core component of the flagship research project of the organization.
 - Developed a tool to screen the results of PubMed queries for relevance to research teams at IHME using natural language processing and deep learning methods.
 - Developed estimates of non-fatal injury burden for the Global Burden of Disease Study 2016 and sexual violence indicators for the Sustainable Development Goals.

Honors & Awards

- 2025 Thomas R. Fleming Excellence in Biostatistics Award
- 2024 [FDA Oncology Center of Excellence & American Statistical Association Educational Fellowship](#)
- 2024 Student Scholarship Award, Biopharmaceutical Section of the American Statistical Association
- 2024 Honorable Mention in the Student Paper Competition, Biopharmaceutical Section of the American Statistical Association
- 2019 [Senior MS in Biostatistics Award](#), Department of Biostatistics, University of Washington
- 2019 Graduate School Conference Travel Award, University of Washington
- 2016 [Husky 100 Award](#), University of Washington
- 2016 [Outstanding Student Award](#), School of Public Health, University of Washington

Software

- 2024, 2025 [RobinCar](#), [RobinCar2](#): Robust Inference for Covariate Adjustment in Randomized Clinical Trials. *These fit-for-purpose R packages unify covariate adjustment methods for randomized clinical trials (they have become the industry standard). RobinCar has 10,000+ downloads from CRAN (600+/-month); RobinCar2 has 2,300+ downloads from CRAN (400+/-month).*

Publications

1. **Bannick MS**, Shao J, Liu J, Du Y, Yi Y, and Ye T. A general form of covariate adjustment in randomized clinical trials under covariate-adaptive randomization. *Biometrika* 3 2025;112. DOI: [10.1093/biomet/asaf029](#).
2. **Bannick MS**, Heltshe S, and Simon N. Accounting for inconsistent use of covariate adjustment in group sequential trials. *Statistics in Medicine* 10-12 2025;44. DOI: [10.1002/sim.70082](#).
3. **Bannick MS**, Donnell D, Hayes R, Laeyendecker O, and Gao F. An enhanced cross-sectional HIV incidence estimator that incorporates prior HIV test results. *Statistics in Medicine* 2024. DOI: [10.1002/sim.10112](#).
4. Ye T, **Bannick MS**, Yi Y, and Shao J. Robust variance estimation for covariate-adjusted unconditional treatment effect in randomized clinical trials with binary outcomes. *Statistical Theory and Related Fields* 2 2023;7:159–63. DOI: [10.1080/24754269.2023.2205802](#).
5. **Bannick MS**, Gao F, Brown E, and Janes H. Retrospective, observational studies for estimating vaccine effects on the secondary attack rate of SARS-CoV-2. *American Journal of Epidemiology* 6 2023;192:1016–28. DOI: [10.1093/aje/kwad046](#).
6. Gao F and **Bannick MS**. Statistical considerations for cross-sectional incidence estimation based on recency test. *Statistics in Medicine* 8 2022;41:1446–61. DOI: [10.1002/sim.9296](#).
7. **Bannick MS**, McGaughey M, and Flaxman A. Ensemble modelling in descriptive epidemiology: burden of disease estimation. *International Journal of Epidemiology* 6 2020;49:2065–74. DOI: [10.1093/ije/dyz223](#).
8. **Norwood MS**, Lupo P, Chow E, Scheurer M, Plon S, Danysh H, Spector L, Carozza S, and Mueller B. Childhood cancer risk in those with chromosomal and non-chromosomal congenital anomalies in Washington State: 1984-2013. *PLoS One* 2017. DOI: [10.1371/journal.pone.0179006](#).

9. **Norwood MS**, Hughes J, and Amico K. The validity of self-reported behaviors: methods for estimating underreporting of risk behaviors. *Annals of Epidemiology* 9 2016;26:612–8. DOI: [10.1016/j.annepidem.2016.07.011](https://doi.org/10.1016/j.annepidem.2016.07.011).

Preprints

10. Pan J, **Bannick MS**, and Gao F. Estimating HIV cross-sectional incidence using recency tests from a non-representative sample. 2024. In revisions at *American Journal of Epidemiology*. DOI: [10.48550/arXiv.2412.12316](https://doi.org/10.48550/arXiv.2412.12316).
11. **Bannick MS** and Simon N. Improved convergence rates of nonparametric penalized regression under misspecified total variation. 2023. DOI: [10.48550/arXiv.2308.01470](https://doi.org/10.48550/arXiv.2308.01470).

Working and Submitted Manuscripts

12. **Bannick, MS**, Yi Y, and Ye T. Covariate adjustment in randomized clinical trials: from general theory to practical insights. 2025+. Under review at *Clinical Trials*.
13. **Bannick, MS**, Bian Y, Chen G, Li L, Qian Y, Sabanés Bove D, Xi D, Ye T, and Yi Y. The RobinCar Family: R tools for robust covariate adjustment in randomized clinical trials. 2025+. In preparation; to be submitted to *BMC Medical Research Methodology (Software)*.
14. **Bannick, MS**, Yi Y, and Ye T. Estimands and estimators of treatment effect in master protocol trials with time-to-event outcomes. 2025+. In preparation.

Book Chapters

15. Van Lancker K, **Bannick, MS**, and Ye T. Covariate Adjustment. In: *Estimands in Clinical Trials: A Practical Guide*. Ed. by Wei J, Meng L, Bretz F, Chen F, and Wang J. *To be released in December 2025*. Springer, 2025.

Global Burden of Disease Collaboration

Author on the following publications as part of the [Global Burden of Disease Study](#). Developed statistical and computational machinery and performed analyses.

16. **GBD 2019 Diseases and Injuries Collaborators**. Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet* 10258 2020;396:1204–22. DOI: [10.1016/S0140-6736\(20\)30925-9](https://doi.org/10.1016/S0140-6736(20)30925-9).
17. **GBD 2016 Neurology Collaborators**. Global, regional, and national burden of neurological disorders, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *The Lancet Neurology* 5 2019;18:459–80. DOI: [10.1016/S1474-4422\(18\)30499-X](https://doi.org/10.1016/S1474-4422(18)30499-X).
18. **GBD 2016 Traumatic Brain Injury and Spinal Cord Injury Collaborators**. Global, regional, and national burden of traumatic brain injury and spinal cord injury, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *The Lancet Neurology* 1 2018;18:56–87. DOI: [10.1016/S1474-4422\(18\)30415-0](https://doi.org/10.1016/S1474-4422(18)30415-0).
19. **GBD 2017 Causes of Death Collaborators**. Global, regional, and national age-sex-specific mortality for 282 causes of death in 195 countries and territories, 1980-2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet* 10159 2018;392:1736–88. DOI: [10.1016/S0140-6736\(18\)32203-7](https://doi.org/10.1016/S0140-6736(18)32203-7).

20. **GBD 2016 Disease and Injury Incidence and Prevalence Collaborators.** Global, regional, and national incidence, prevalence, and years lived with disability for 328 diseases and injuries for 195 countries, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *The Lancet* 10100 2017;390:1211–59. DOI: [10.1016/S0140-6736\(17\)32154-2](https://doi.org/10.1016/S0140-6736(17)32154-2).
21. **GBD 2016 DALYs and HALE Collaborators.** Global, regional, and national disability-adjusted life-years for 332 diseases and injuries and healthy life expectancy (HALE) for 195 countries and territories, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *The Lancet* 10100 2017;390:1260–344. DOI: [10.1016/S0140-6736\(17\)32130-X](https://doi.org/10.1016/S0140-6736(17)32130-X).
22. **GBD 2016 SDG Collaborators.** Measuring progress and projecting attainment on the basis of past trends of the health-related Sustainable Development Goals in 188 countries: an analysis from the Global Burden of Disease Study 2016. *The Lancet* 10100 2017;390:1423–59. DOI: [10.1016/S0140-6736\(17\)32336-X](https://doi.org/10.1016/S0140-6736(17)32336-X).

Presentations

23. **Bannick, MS.** Estimands and estimators for time-to-event outcomes in platform trials. WNAR Conference. Oral Presentation. Whistler, British Columbia, 2025.
24. **Bannick, MS.** A novel covariate adjustment strategy for guaranteed efficiency gain in randomized clinical trials. Computational and Methodological Statistics Conference. Oral Presentation. Berlin, Germany (remote), 2023.
25. **Bannick, MS.** Accounting for inconsistent use of covariate adjustment in group sequential trials. ENAR Conference. Oral Presentation. Nashville, Tennessee, 2023.
26. **Bannick, MS.** Estimating time to intermediate endpoints using population-level survival data and deconvolution methods, with application to cancer progression and recurrence. Women in Statistics and Data Science Conference. [Oral and Poster Presentation](#). Bellevue, Washington, 2019.
27. **Bannick, MS.** Estimating time to intermediate endpoints using population-level survival data and deconvolution methods, with application to cancer progression and recurrence. Joint Statistical Meetings. [Poster Presentation](#). Denver, Colorado, 2019.
28. **Bannick, MS.** Behind the Scenes: Building Tools to Visualize Intermediate Results in Complex Data Science Pipelines. Symposium on Data Science and Statistics. [Invited Presentation](#). Bellevue, Washington, 2019.
29. **Bannick, MS.** Cause of Death Modeling. Global Burden of Disease Technical Workshop. Plenary Session. Eretria, Greece, 2019.
30. Misganaw A, **Bannick, MS**, and Srinivasan V. Ethiopia Disease Burden within the Global Burden of Disease Study 2016. Ethiopian Public Health Institute. Addis Ababa, Ethiopia, 2018.
31. **Bannick, MS.** Childhood cancer in relation to the presence of congenital malformations in Washington State. School of Public Health Undergraduate Symposium. Poster Presentation. University of Washington, Seattle, Washington, 2016.
32. **Bannick, MS.** The Validity of Self-Reported Behaviors: Methods for Estimating Underreporting of Risky Behaviors. School of Public Health Undergraduate Symposium. Poster Presentation. University of Washington, Seattle, Washington, 2015.

33. **Bannick, MS.** A Public Health Approach to Parenting Interventions: Implementation Issues. School of Public Health Undergraduate Symposium. Poster Presentation. University of Washington, Seattle, Washington, 2014.

Service and Affiliations

Affiliations

2019–present American Statistical Association (ASA)

Peer Review

2025 Clinical Trials

2025 The American Statistician

2025 Statistics in Biopharmaceutical Research

2024 Pharmaceutical Statistics

2021 International Journal of Epidemiology

2020, 2021 Machine Learning in Public Health Workshop, NeurIPS

2020 Journal of Medical Internet Research

Committees

2024–2025 Software Subteam, ASA Biopharmaceutical Section Covariate Adjustment Scientific Working Group

2023 Student Representative, UW Biostatistics

2022 Graduate Program Admissions, UW Biostatistics

2021 Open-rank Faculty Search Committee, UW Biostatistics

2020, 2021 Equity, Diversity, and Inclusion Committee, UW Biostatistics

2020, 2021 Educational Policy and Teaching Evaluation Committee, UW Biostatistics

Teaching

Teaching Assistant

Winter 2021 BIOST/EPI 537: Survival Data Analysis in Epidemiology, for Prof Jon Wakefield, University of Washington, Seattle. <https://github.com/mbannick/survival-discussion-section>

Spring 2021 BIOST/STAT 524: Design of Medical Studies, for Prof Thomas Fleming, University of Washington, Seattle.

07/2021 Supervised Methods for Statistical Machine Learning, for Profs Ali Shojaie and Noah Simon, Summer Institute in Statistics for Big Data, University of Washington.

05/2022 Workshop on Statistical Machine Learning, for Prof Noah Simon, ICES Ontario (remote).

07/2022 Causal Inference with Observational Data: Common Designs and Statistical Methods, for Profs Ting Ye and Qingyuan Zhao, Summer Institute in Statistics for Clinical & Epidemiological Research, University of Washington.

07/2023 Special Topics in Clinical Trials: Interpreting Exploratory Analyses & When are Biomarkers Valid Surrogates? for Prof Thomas Fleming, Summer Institute in Statistics for Clinical & Epidemiological Research, University of Washington.

Guest Lectures

04/2023 & 2025 "Selected Design Issues", Design of Medical Studies, *Instructor: Dr. Thomas Fleming*, University of Washington, Seattle

- 07/2020 "Introduction to Epidemiological and Biostatistical Thinking", Neurology Clinical Fellowship Didactics *Instructor: Dr. Andrea Cheng-Hakimian*, University of Washington, Seattle.
<https://github.com/mbannick/uw-neurology-fellows>
- 08/2018 & 2019 "Cause of Death Ensemble Model (CODEm)", Global Burden of Disease (GH 590) *Instructor: Dr. Jeffrey Stanaway*, Department of Global Health, University of Washington, Seattle

Workshops

- 09/2020 "Introduction to Research at the Institute for Health Metrics and Evaluation: Training Bootcamp for First-Year Post-Bachelor Fellows". Designed curriculum and facilitated a week-long crash course on intro epidemiology, biostatistics, R, Git and high performance computing. Institute for Health Metrics and Evaluation, University of Washington, Seattle.
- 05/2019 "Data to DALYs: Case Study on Diabetes", with Dr. Theo Vos and Dr. Liane Ong. 2-day short course. Global Burden of Disease Workshop, Eretria, Greece.

Tutoring

- Fall 2020 Medical Biometry I (BIOST 511), University of Washington, Seattle

Additional Training and Research

Pre-Graduate Research

- 2015–2016 **Research Assistant**, *Fred Hutchinson Cancer Research Center*.
- Performed statistical analyses for a cohort study of pregnancy outcomes in women with multiple sclerosis, and a [case-control study of congenital malformations and childhood cancer](#).
 - Researched the capacity of each state in the U.S. to link birth certificates to state cancer registries for a National Cancer Institute-funded study.
- 2015 **Research Assistant**, *Department of Biostatistics, University of Washington*.
- Developed a statistical method to estimate under-reporting of sensitive, self-reported behaviors in a study population with biomarkers.
 - [Authored a publication on the novel method](#) that was presented by Dr. James Hughes at the CDC Expert Consultation on Advancing Methods for Biobehavioral Surveys in 2018.
- 2014 **Research Assistant**, *Fred Hutchinson Cancer Research Center*.
HIV Prevention Trials Network
- Developed an R program for an HIV Prevention Trials Network study to inform the categorization of biological specimens in a way that optimized sensitivity and specificity.
- 2014 **Undergraduate Research Assistant**, *University of Washington*.
Center for Clinical and Epidemiological Research
- Supported the maintenance of a large health research registry.
 - Performed targeted literature reviews to inform grants for new epidemiological twin studies.
- 2013–2014 **Undergraduate Research Assistant**, *University of Washington*.
Division of Public Behavioral Health and Justice Policy
- Analyzed qualitative survey data using ATLAS.ti to determine barriers to implementing evidence-based parenting interventions in Washington State.
 - Designed online data collection platforms for intervention monitoring and evaluation.
 - Assisted in writing monitoring and evaluation progress reports for the Washington State Division of Behavioral Health and Recovery.

Additional Training

- 2015 Summer Institute in Statistics and Modeling in Infectious Diseases, University of Washington
- 2015 Writing in the Sciences, *with distinction*, via Stanford Online, Lagunita